Building Circuits – Week 8

1. LED in a circuit

An LED (light emitting diode) is a component that only allows current to flow in one direction, and it emits light when current passes through it. Like all diodes, it has a positive side (called the anode, the longer wire) and a negative side (called the cathode, the shorter wire).

a. Build a circuit with a resistor of at least 150Ω and an LED in series, connected between the +5V and ground rows. ✅

Used 220 Ohm resistor

b. How does the diode need to be oriented? Which wire on the LED goes to the +5V side and which goes to the GND connector?

The diode should be oriented in forward bias, with the anode (positive side) of the LED connected to the +5V side and the cathode (negative side) of the LED connected to GND

c. What is the voltage drop across the resistor? Was this what you expected?

The measured voltage drop across the resistor is 2.11 V. The expected voltage drop is 5V - 3.2V = 1.8V.

d. What is the voltage drop across the LED?

The measured voltage drop across the LED is 2.84 V. The expected voltage drop is 3.2 V.

e. Try removing the resistor from the circuit, keeping the circuit closed – the LED is just in series with the 5V supply. ✅

f. What do you think will happen to the LED brightness?

It will get brighter (and might burn out if left in this configuration for too long).

NOTE: Do not leave the LED in this circuit closed for very long.

g. What important purpose does the resistor serve?

The resistor serves the purpose of dropping the voltage across the LED so the circuit isn't shorted and the LED isn't overloaded.

h. Try including resistors of different values – how does LED brightness change vs resistor strength?

As resistance increases, LED brightness decreases.

i. Measure the voltage drop across the resistor and LED in each case and make note of voltage drop versus resistance.

R = 150 Ω

Experimental: V = 1.95 V across the resistor

(Brighter than the light emitted by the LED when the 220 Ω resistor was used)

R = 220 Ω

Experimental: V = 2.11 V

R = 47,000 Ω

Experimental: V = 2.55 V

(Very, very dim)

j. Using the configuration with the highest LED brightness, move the 5V connection on the RPi to one of the 3.3V pins. What do you expect to happen to the LED brightness?

   LED brightness gets significantly dimmer as it is supplied a lower voltage.

k. Move the RPi voltage connection back to 5V. ✅

2. Boost converter (step-up voltage)

A boost converter is a small circuit that takes a lower input voltage and converts it to a higher output voltage. You will use one to take the 5V from the RPi and increase it. This is the same kind of component we will use in a future lab to provide the higher voltage needed to power a radiation detector.

l. Add your step-up circuit component to increase your RPi voltage from 5V. Do not close your circuit yet.✅

m. Use the voltmeter to examine the output voltage of the step-up component and dial it up or down until the output is above 5V but below 10V.

   9.85 V

n. Using the dimmest LED configuration from before (meaning select the appropriate resistor from those you tried previously), connect the LED circuit to the boost converter output instead of the RPi 5V directly. ✅

o. How does the LED brightness change as you increase the output voltage?

   As the output voltage increases, the voltage gets a little brighter (the resistor is very large).

p. How does the voltage drop across the resistor and LED change?

   As the output voltage increases, the voltage drop across the resistor was about 8.3 V (increased) but the voltage drop across the LED remains constant (fixed property of the LED).

q. Do any of these results change with different color LEDs? Specifically, do any voltage drop values change? Is the relative brightness similar for different color LEDs?

   Yellow: 2 - 2.2 V

   - **Voltage across LED: 1.8 V**
   - **Voltage across resistor: 9.07 V**

   Green (same V as blue): 3 - 3.2 V

   - **Voltage across LED: 2.2 V (lower than what we should be measuring, but the blue was also rated 3 - 3.2 V and we measured around 2.5 V)**
   - **Voltage across resistor: 8.66 V (difference between supply + drop)**

   NOTE: The boost converter works by rapidly switching current on and off internally. This creates a steady higher voltage on the output, but it also produces small, fast fluctuations called "ripple" or "noise" riding on top of that steady voltage. For powering an LED this does not matter, but in a future lab when we use a boost converter to power a sensitive detector, this noise will become very important. Keep this in the back of your mind.

3. Photo-diode

A photo-diode is a type of diode that generates a small electrical current when light hits it. Unlike the LED (which produces light from current), the photo-diode does the opposite – it converts light into current. Photo-diodes are used in "reverse bias" mode, meaning you connect the voltage backwards compared to how you connected the LED. In reverse bias, very little current flows normally, but when light hits the diode, it creates a small current that is proportional to the amount of light.

a. Replace the LED with a photo-diode. Start with the step-up component outputting 5V. ✅ **(4.87 V, couldn't get exactly to 5 V)**

   NOTE: Photo-diodes operate in reverse bias mode so you will need to orient the diode the opposite way from how you had the LED.

b. What is the voltage across the resistor when you have a 5V bias and the photo-diode is exposed to the ambient light in the room?

   4.28 V

c. What happens if you cover the photo-diode with your hand? What happens if you increase the bias voltage to 10V?

   Cover with hand: Originally around 4.28 V when the photodiode is covered with my hand, the voltage across the resistor drops to 0.027 V (not exactly zero).

d. What is the "dark current" for this photo-diode? Dark current is the tiny amount of current that flows through the diode even when no light is hitting it. Use the voltage across the resistor and Ohm's law to calculate the current flowing through the diode.

   Dark current is caused by electrons becoming excited by thermal energy, which, even in the absence of light can still be excited and show a small voltage.

   The dark current is I = V / R = 0.027 V / 47,000 Ω = 5.74 * 10^-7 amps

e. Is 5V enough supply voltage to see a signal from this diode?

   Yes; there's a voltage drop of around 4 V across the resistor, so there must be current and a signal from the diode: I = V / R = 4V / 47,000 Ω = 8.51 * 10^-5 amps

f. Try shining the flashlight from your phone onto the diode. What is the current output? Again, use the voltage across the resistor to determine this.

   The voltage across the resistor increased to 4.72 V. The current would be I = V/R = 4.72 V / 47,000 Ω = 0.0001 A

g. Are you able to determine the saturation current for the photo-diode? Saturation current is the maximum current the diode will produce no matter how much more light you add.

   Using a phone flashlight, we can start really far from the photodiode, and keep moving it closer until it stops going up (if it ever stops going up).

   Experimental: The voltage across the resistor was capped at around 4.72 V, which gives a saturation current of 0.0001 A, but this voltage is also very close to the voltage supplied by the boost converter, so this may not actually be the saturation current.

4. Looking ahead

The photo-diode you just worked with produces a current when it detects light. Next week, you will work with a similar kind of sensor – a PIN diode – that is sensitive enough to detect individual flashes of light produced by radiation interacting with a crystal. The key challenge you will face is that the signal from a bare diode like this is extremely small. You will use an amplifier circuit to make that signal large enough to see on an oscilloscope. You will also use a boost converter like the one from today’s lab to power a radiation detector, and the filtering concepts from last week’s simulation will become directly relevant for getting a clean signal.